

LINEAR REGRESSION: Predicting Revenue from Advertising Spend

Scenario: You are part of a smartwatch startup. Your marketing team has been running advertisements across multiple channels (social media, TV, print). The CFO wants to predict revenue for the next month based on ad spend data. Your task is to use linear regression to predict revenue and analyze the impact of increasing the advertising budget.

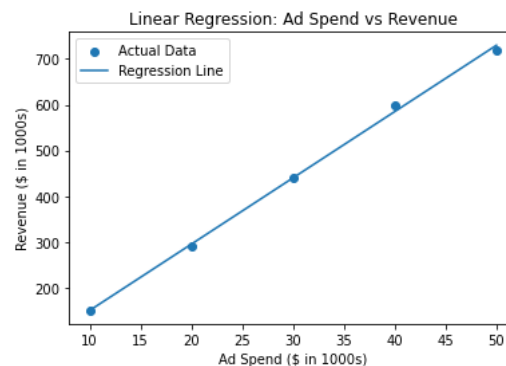
Question:

You have the following data:

Ad Spend vs Revenue Data

Ad Spend (\$ in 1000s)	Revenue (\$ in 1000s)
10	150
20	290
30	440
40	600
50	720

Use linear regression to find the relationship between ad spend and revenue.



What is the expected revenue if the ad spend is \$60,000?

When ad spend is \$60,000, the expected revenue is \$875,000.

What does the slope of the line represent in this context?

In general, the slope of a line represents the rate of change between two variables. In this case, the slope represents the positive linear relationship between ad spend and revenue. For every \$1,000 in ad spend, revenue increases by \$14,500. This does not prove that ad spend causes revenue to increase; it simply illustrates the association between the two.

LOGISTIC REGRESSION: Predicting Subscription Renewal

Scenario: You are managing a fitness subscription platform. The company wants to predict whether users will renew their subscriptions based on their activity levels. The dataset contains user activity data and renewal status.

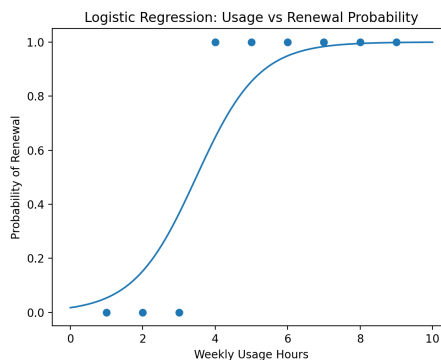
Question:

Weekly Usage vs. Renewal Status

Weekly Usage Hours	Renewal Status (1 = Yes, 0 = No)
1	0
3	0
5	1
7	1
9	1

Use logistic regression to model the probability of renewal based on weekly usage hours.

Included additional usage hour data points (2, 4, 6, 8) to visually confirm my responses below.



What is the probability that a user logging 4 hours per week will renew their subscription?

The probability that a user logging 4 hours per week renews their membership is 0.5043, or 50%.

Based on the model, recommend a threshold for targeting at-risk users with promotions.

Based on the model and a 0.5 decision line, I recommend targeting users logging 4 hours or fewer with promotions.

MODEL SELECTION: Choosing Between Linear & Logistic Regression

Scenario: You are part of a data science team at a health-tech startup. The company is launching two initiatives:

Predicting Patient Recovery Time: Based on historical data, the company wants to predict the number of days it takes for a patient to recover from surgery.

Predicting Disease Diagnosis: The company also wants to predict whether a patient has a specific disease based on diagnostic test results (positive/negative).

Your task is to decide which regression model—linear or logistic regression—is appropriate for each initiative, justify the choice, and explain the differences between these methods.

Question:

Which regression model (linear or logistic) is suitable for each of the two initiatives? Justify your choice.

Linear Regression is suitable for predicting patient recovery time because it is designed to model continuous values, in this case, the number of recovery days. Logistic Regression is suitable for predicting disease diagnosis because the outcome is binary (positive or negative) and estimates the probability of the disease.

Explain the key differences between linear and logistic regression, focusing on:

- The type of dependent variable they model.
Linear Regression – Continuous / Numeric values
Logistic Regression – Binary values (0/1, yes/no, positive/negative)
- The mathematical equations used.
Linear Regression - $y = \beta_0 + \beta_1 x_1$ (produces a straight line)
Logistic Regression - $P(y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 x)})$ (produces an s-curve)
- Real-world scenarios where each is used.
Linear Regression predicts home prices, sales revenue, investment risk, and demand forecasting
Logistic Regression predicts fraud detection, employee attrition risk, and loan approval

Assume you are building a model for Disease Diagnosis. Based on the logistic regression equation: Calculate the probability of a disease diagnosis for a patient with a test score $x = 5$, given $\beta_0 = -2$ and $\beta_1 = 0.8$.

Based on the logistic regression equation, a patient with a test score of 5 has about an 88.1% chance of having the disease.